

ADMM Algorithm for Nonconvex Problems: Applications and Convergence Analysis

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Sept. 2014

Agenda

- Review a few recent applications of ADMM on nonconvex problems
- A new analysis for nonconvex ADMM
 - ① Basic proof steps
 - ② Extensions
- Road map ahead

The Basic Setup

- Consider the following linearly constrained (possibly **nonsmooth** or/and **nonconvex**) problem with K blocks of variables $\{x_k\}_{k=1}^K$:

$$\begin{aligned} \min \quad & f(x) := \sum_{k=1}^K g_k(x_k) + \ell(x_1, \dots, x_K) \\ \text{s.t.} \quad & \sum_{k=1}^K A_k x_k = q, \quad x_k \in X_k, \quad \forall k = 1, \dots, K \end{aligned} \tag{1}$$

- Here $A_k \in \mathbb{R}^{M \times N_k}$ and $q \in \mathbb{R}^M$; $X_k \in \mathbb{R}^{N_k}$ is a closed convex set; $\ell(\cdot)$ is a smooth (possibly nonconvex) function; each $g_k(\cdot)$ can be either a smooth function, or a convex nonsmooth function
- Define $A := [A_1, \dots, A_K]$

The Basic Setup

- The augmented Lagrangian for problem (1) is given by

$$L(x; y) = \sum_{k=1}^K g_k(x_k) + \ell(x_1, \dots, x_K) + \langle y, q - Ax \rangle + \frac{\rho}{2} \|q - Ax\|^2,$$

where $\rho > 0$ is a constant representing the primal stepsize.

Algorithm 0. ADMM for Problem (1)

At each iteration $t + 1$, update the primal variables:

$$x_k^{t+1} = \arg \min_{x_k \in X_k} L(x_1^{t+1}, \dots, x_{k-1}^{t+1}, x_k, x_{k+1}^t, \dots, x_K^t; y^t), \quad \forall k. \quad (2)$$

Update the dual variable:

$$y^{t+1} = y^t + \rho(q - Ax^{t+1}). \quad (3)$$

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Overview of Applications

- ADMM has been widely (and wildly) used to solve nonconvex problems
 - 1 nonnegative matrix factorization [Sun-Fevotte 2014]
 - 2 phase retrieval [Wen-Yang-Liu-Marchesini 12]
 - 3 distributed matrix factorization [Ling-Yin-Wen 12]
 - 4 distributed clustering [Forero-Cano-Giannakis 11]
 - 5 polynomial optimization/tensor decomposition [Jiang-Ma-Zhang 13, Livavas-Sidiropoulos 14]
 - 6 matrix separation/completion [Xu-Yin-Wen-Zhang 11]
 - 7 asset allocation [Wen-Peng-Liu-Bai-Sun 13]
- Even for discontinuous problems
- However, no formal convergence analysis is known
- Most analysis depends on **uncheckable conditions** such that $\|x^{t+1} - x^t\| \rightarrow 0$, $\|y^{t+1} - y^t\| \rightarrow 0$, etc.

Application 1: NMF

- Many papers, when justifying their use of nonconvex ADMM, cite Chapter 9 of the book by Boyd *et al* 2011
- In this chapter, a few nonconvex problems are discussed
- The only continuous problem discussed there is the following nonnegative matrix factorization

$$\min \frac{1}{2} \|VW - C\|_F^2, \text{ s.t. } V_{ij} \geq 0, W_{ij} \geq 0 \quad (4)$$

- Introduce a new variable Z , we have the equivalent form

$$\min \frac{1}{2} \|Z - C\|_F^2, \text{ s.t. } Z - VW = 0, V_{ij} \geq 0, W_{ij} \geq 0 \quad (5)$$

Application 1: NMF (cont.)

- The ADMM iteration is given by

$$(Z^{t+1}, V^{t+1}) = \arg \min_{Z, V \geq 0} (\|Z - C\|_F^2 + (\rho/2)\|Z - VW^t + U^t\|)$$

$$W^{t+1} := \arg \min_{W \geq 0} \|Z^{t+1} - V^{t+1}W + U^t\|_F^2$$

$$U^{t+1} := U^t + Z^{t+1} - V^{t+1}W^{t+1}$$

- “The first step splits across the rows of Z and V , so can be performed by solving a set of quadratic programs, in parallel”
- “The second splits in the columns of W , so can be performed by solving parallel quadratic programs to find each column”
- Subproblems are NOT in closed form
- No convergence analysis

Application 2: NMF with Missing Values

- This formulation comes from [Xu-Yin-Wen-Zhang 11]
- Consider the following problem

$$\min_{X,Y} \|P_{\Omega}(XY - M)\|_F^2 \quad \text{s.t. } X_{ij} \geq 0, Y_{ij} \geq 0 \quad (6)$$

- Ω indexes the known entries of M
- P_{Ω} is the projection to Ω
- Introduce **three** new variables Z, U, V (note X, Y are **coupled** in the objective)

$$\begin{aligned} \min_{X,Y,Z,U,V} \|XY - Z\|_F^2 \\ \text{s.t. } U_{ij} \geq 0, V_{ij} \geq 0, \quad X = U, Y = V, \\ P_{\Omega}(Z - M) = 0 \end{aligned}$$

- Introduce multipliers Λ (for U) and Π (for V)

Application 2: NMF with Missing Values (cont.)

- All the primal updates can then be written down in closed-form

$$X^{t+1} = (Z^t(Y^t) + \alpha U^t - \Lambda^t)(Y^t(Y^t)^T + \alpha I)^{-1}$$

$$Y^{t+1} = \left((X^{t+1})^T X^{t+1} + \beta I \right)^{-1} \left((X^{t+1})^T Z^t + \beta V^t - \Pi^t \right)$$

$$Z^{t+1} = X^{t+1} Y^{t+1} + P_{\Omega} (M - X^{t+1} Y^{t+1})$$

$$U^{t+1} = P_+ (X^{t+1} + \Omega^t / \alpha)$$

$$V^{t+1} = P_+ (Y^{t+1} + \Pi^t / \beta)$$

- Convergence analysis** If the successive differences of all the primal and dual variables go to zero (e.g., $X^{t+1} - X^t \rightarrow 0$ and so on), then any accumulation point of the iterates is a KKT point of the reformulated problem
- This algorithm has been extended in [Sun-Fevotte 14] to **general β -divergence criteria** for NMF; Also extended to **Tensor Decomposition** [Liavas-Sidiropoulos 14]; Again closed-form updates, no convergence analysis

Application 3: Matrix Low Rank Factorization

- Consider the following formulation of low-rank+sparse matrix factorization

$$\min_{U,V} \|P_{\Omega}(UV - D)\|_1 \quad (7)$$

where $U \in \mathbb{R}^{m \times k}$ and $V \in \mathbb{R}^{k \times n}$ and k is small

- Motivation:** Avoid nuclear norm minimization; assume the knowledge that the rank of the desired low rank matrix is less than k
- Write down the equivalent formulation

$$\min_{U,V,Z} \|P_{\Omega}(Z - D)\|_1, \text{ s.t. } UV - Z = 0 \quad (8)$$

- Again closed form solution; convergence based on assumption on diminishing distance between iterates

Application 3: Matrix Low Rank Factorization (cont.)

- A related algorithm tries to compute a matrix low-rank factorization in a decentralized way [Zhang-Kowk 14]
- Consider the matrix low-rank factorization problem

$$\min_{X,Y} = \frac{1}{2} \|UV - D\|_F^2 = \sum_{m=1}^M \|UV(m) - D(m)\|_F^2 \quad (9)$$

where $V(m)$ and $D(m)$ is the m th column of V and D

- Each $V(m)$ and $D(m)$ is **private**, only available to a single agent m
- Equivalently

$$\min_{U,V,Z} = \sum_{m=1}^M \|Z_m V(m) - D(m)\|_F^2, \text{ s.t. } Z_m = U, \forall m \quad (10)$$

Application 4: Finding Intersection of Two Sets

- This type of applications tries to find the intersection of two (possibly nonconvex sets) [Wen-Yang-Liu-Marchesini-12]
- Problem Statement

$$\text{find } x \text{ and } y, \quad \text{s.t. } x = y, \quad x \in X, \quad y \in Y. \quad (11)$$

- ADMM iteration is very simple

$$\begin{aligned} x^{t+1} &= P_X (y^t - \lambda^t) \\ y^{t+1} &= P_Y (x^{t+1} + \lambda^t) \end{aligned}$$

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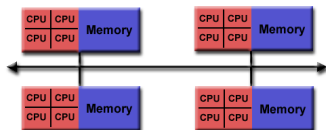
Problem Setup: A nonconvex Consensus Problem

- Consider a nonconvex global consensus problem

$$\min \quad f(x) := \sum_{k=1}^K g_k(x) + h(x) \quad \text{s.t.} \quad x \in X \quad (12)$$

- g_k : smooth, possibly **nonconvex** function
- $h(x)$: convex nonsmooth regularization term
- Each g_k needs to be handled by a single agent

$$\min \sum_{k=1}^K g_k(x_k) + h(x) \quad \text{s.t.} \quad x_k = x, \quad \forall k = 1, \dots, K, \quad x \in X. \quad (13)$$



Application: Distributed Sparse-PCA

- The sparse PCA problem can be formulated as

$$\min_x x^T B x + \gamma \|x\|_0, \quad \|x\|_2^2 \leq 1 \quad (14)$$

where $-B \succ 0$

- Consider its relaxation

$$\min_x x^T B x + \gamma \|x\|_1, \quad \|x\|_2^2 \leq 1 \quad (15)$$

- Has wide applications, for example large-scale text data analysis

Application: Distributed Sparse-PCA (cont.)

- In large-scale text analysis, let $C \in \mathbb{R}^{R \times N}$ denote the summary of the text corpora such that
 - A total of R documents (one row one document); N words
 - $C[r, n] = 1$ means word n appears in document r
- Standard sparse PCA analysis

$$\min_x -x^T C^T C x + \gamma \|x\|_1, \quad \|x\|_2^2 \leq 1 \quad (16)$$

- Suppose data is divided by $C = [C_1^T, \dots, C_K^T]^T$, where each $C_k \in \mathbb{R}^{r_k \times N}$ consists of a subset of documents
- C_k 's stored on K different agents; no data exchange allowed

$$\min_x - \sum_{k=1}^K x^T C_k^T C_k x + \gamma \|x\|_1, \quad \|x\|_2^2 \leq 1 \quad (17)$$

- A nonconvex global consensus problem

The Algorithm

- The augmented Lagrangian function is given by

$$L(\{x_k\}, x; y) = \sum_{k=1}^K g_k(x_k) + h(x) + \sum_{k=1}^K \langle y_k, x_k - x \rangle + \sum_{k=1}^K \frac{\rho_k}{2} \|x_k - x\|^2.$$

Algorithm 1. The Classical ADMM for the Consensus Problem (13)

At each iteration $t + 1$, compute:

$$x^{t+1} = \underset{x \in X}{\operatorname{argmin}} L(\{x_k^t\}, x; y^t) = \operatorname{prox}_{l(X)+h} \left[\frac{\sum_{k=1}^K \rho_k x_k^t + \sum_{k=1}^K y_k^t}{\sum_{k=1}^K \rho_k} \right].$$

Each node k computes x_k by solving:

$$x_k^{t+1} = \arg \min_{x_k} g_k(x_k) + \langle y_k^t, x_k - x^{t+1} \rangle + \frac{\rho_k}{2} \|x_k - x^{t+1}\|^2.$$

Each node k updates the dual variable:

$$y_k^{t+1} = y_k^t + \rho_k (x_k^{t+1} - x^{t+1}).$$

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Assumptions

Assumption A.

A1. g_k 's Lipschitz continuous:

$$\|\nabla_k g_k(x_k) - \nabla_k g_k(z_k)\| \leq L_k \|x_k - z_k\|, \quad \forall x_k, z_k, k = 1, \dots, K.$$

Moreover, h is convex (possibly nonsmooth); X is a closed convex set.

A2. ρ_k is large enough such that:

- 1 For all k , the x_k subproblem is strongly convex with modulus $\gamma_k(\rho_k)$;
- 2 For all k , $\rho_k \gamma_k(\rho_k) > 2L_k^2$ and $\rho_k \geq L_k$.

A3. $f(x)$ is bounded from below over X .

Comments:

- As ρ_k increases, x_k subproblem eventually becomes strongly convex
- By construction, the x subproblem is also strongly convex
- No assumption on the iterates generated by the algorithm

- Use $L(x, \{x_k\}; y)$ as the **potential function**
- Use a three-step approach
- **Step 1:** Sufficient Descent

$$\begin{aligned} & L(x^{t+1}, \{x_k^{t+1}\}; y^{t+1}) - L(x^t, \{x_k^t\}; y^t) \\ & \leq -\sigma_0 \|x^{t+1} - x^t\|^2 - \sum_{k=1}^K \sigma_k \|x_k^{t+1} - x_k^t\|^2 \end{aligned}$$

- **Step 2:** Show that $L(x^{t+1}, \{x_k^{t+1}\}; y^{t+1})$ is lower bounded
- **Step 3:** Show that $\{x^{t+1}, \{x_k^{t+1}\}, y^{t+1}\}$ converges to a **stationary solution** of the global consensus problem

S1: Sufficient Descent

- To show sufficient descent, we need the following lemma
- **Lemma:** The following is true

$$L_k^2 \|x_k^{t+1} - x_k^t\|^2 \geq \|y_k^{t+1} - y_k^t\|^2, \forall k = 1, \dots, K, \quad (18)$$

- From the x_k update step, we have the following optimality condition

$$\nabla g_k(x_k^{t+1}) + y_k^t + \rho_k(x_k^{t+1} - x_k^t) = 0, \forall k. \quad (19)$$

which implies

$$\nabla g_k(x_k^{t+1}) = -y_k^{t+1}, \forall k. \quad (20)$$

- Using the Lipschitz continuity assumption on ∇g_k we are done

S1: Sufficient Descent (cont.)

- Using the previous lemma, we can show the following

$$\begin{aligned} & L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^t\}, x^t; y^t) \\ & \leq \sum_{k=1}^K K \left(\frac{L_k^2}{\rho_k} - \frac{\gamma_k(\rho_k)}{2} \right) \|x_k^{t+1} - x_k^t\|^2 - \frac{\gamma}{2} \|x^{t+1} - x^t\|^2. \end{aligned} \quad (21)$$

where $\gamma = \sum_{k=1}^K \rho_k$

- First split the successive difference of the augmented Lagrangian by

$$\begin{aligned} & L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^t\}, x^t; y^t) \\ & = \left(L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^{t+1}\}, x^{t+1}; y^t) \right) \\ & \quad + \left(L(\{x_k^{t+1}\}, x^{t+1}; y^t) - L(\{x_k^t\}, x^t; y^t) \right) \end{aligned} \quad (22)$$

S1: Sufficient Descent (cont.)

- The red term can be bounded by

$$\begin{aligned} & L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^{t+1}\}, x^{t+1}; y^t) \\ &= \sum_{k=1}^K \langle y_k^{t+1} - y_k^t, x_k^{t+1} - x^{t+1} \rangle = \sum_k \rho_k \|x_k^{t+1} - x^{t+1}\|^2 \\ &= \sum_k \frac{1}{\rho_k} \|y_k^{t+1} - y_k^t\|^2 \leq \sum_k \frac{L_k^2}{\rho_k} \|x_k^{t+1} - x_k^t\|^2 \end{aligned} \quad (23)$$

- The blue term can be bounded by (using per-block strong convexity)

$$\begin{aligned} & L(\{x_k^{t+1}\}, x^{t+1}; y^t) - L(\{x_k^t\}, x^t; y^t) \\ &= L(\{x_k^{t+1}\}, x^{t+1}; y^t) - L(\{x_k^t\}, x^{t+1}; y^t) \\ &\quad + L(\{x_k^t\}, x^{t+1}; y^t) - L(\{x_k^t\}, x^t; y^t) \\ &\leq - \sum_k \frac{\gamma_k(\rho_k)}{2} \|x_k^{t+1} - x_k^t\|^2 - \frac{\gamma}{2} \|x^{t+1} - x^t\|^2, \end{aligned} \quad (24)$$

S2: Lower Bounds

- We then show that for some constant L^*

$$\lim_{t \rightarrow \infty} L(\{\{x_k^t\}, x^t, y^t\} = L^* > -\infty \quad (25)$$

- We have the following inequalities

$$\begin{aligned} & L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) \\ &= h(x^{t+1}) + \sum_{k=1}^K g_k(x_k^{t+1}) + \langle y_k^{t+1}, x_k^{t+1} - x^{t+1} \rangle + \frac{\rho_k}{2} \|x_k^{t+1} - x^{t+1}\|^2 \\ &= h(x^{t+1}) + \sum_{k=1}^K g_k(x_k^{t+1}) + \langle \nabla g_k(x_k^{t+1}), x^{t+1} - x_k^{t+1} \rangle + \frac{\rho_k}{2} \|x_k^{t+1} - x^{t+1}\|^2 \\ &\stackrel{(a)}{\geq} h(x^{t+1}) + \sum_{k=1}^K g_k(x^{t+1}) = f(x^{t+1}) \end{aligned} \quad (26)$$

where (a) comes from the Lipschitz continuity assumption, and the fact that $\rho_k \geq L_k$ for all $k = 1, \dots, K$

- By assumption A3, $f(x)$ is lower bounded over $x \in X$

S3: Convergence

- **Theorem** Assume that Assumption A is satisfied. Then we have the following
 - ① We have $\lim_{t \rightarrow \infty} \|x_k^{t+1} - x^{t+1}\| = 0, k = 1, \dots, K$
 - ② Let $(\{x_k^*\}, x^*, y^*)$ denote **any limit point** of the sequence $\{\{x_k^{t+1}\}, x^{t+1}, y^{t+1}\}$ generated by Algorithm 1, then it is a stationary solution of problem (13)
 - ③ If **X is a compact set**, then the sequence of iterates generated by Algorithm 1 converges to **the set of stationary solutions** of problem (13)

S3: Convergence (cont.)

- To show the first item, notice that

$$\begin{aligned} & L(\{x_k^{t+1}\}, x^{t+1}; y^{t+1}) - L(\{x_k^t\}, x^t; y^t) \\ & \leq \sum_k \left(\frac{L_k^2}{\rho_k} - \frac{\gamma_k(\rho_k)}{2} \right) \|x_k^{t+1} - x_k^t\|^2 - \frac{\gamma}{2} \|x^{t+1} - x^t\|^2 \end{aligned}$$

- So the lower boundedness of L implies that

$$\|x^{t+1} - x^t\| \rightarrow 0, \quad \|x_k^{t+1} - x_k^t\| \rightarrow 0, \quad \forall k = 1, \dots, K. \quad (27)$$

- By the Lemma, we further obtain $\|y_k^{t+1} - y_k^t\| \rightarrow 0$ for all $k = 1, 2, \dots, K$, which implies that $\|x_k^{t+1} - x^{t+1}\| \rightarrow 0$
- The rest of the proof is checking KKT condition, omitted

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Extensions to Flexible Update Rules

- The above basic analysis can be extended to an implementation with **flexible** update rules
- Let $k = 1, 2, \dots, K$ be the indices for the primal variable blocks $x_1, x_2, \dots, x_k$ and $k = 0$ be the index for primal variable block x
- Let $\mathcal{C}^t \subseteq \{0, 1, \dots, K\}$ denote the set of variables updated in iteration t , then
 - ① **Randomized update rule:** At each iteration $t + 1$, a variable block k is chosen randomly with probability p_k^{t+1} ,

$$\Pr(k \in \mathcal{C}^{t+1} \mid x^t, y^t, \{x_k^t\}) = p_k^{t+1} \geq p_{\min} > 0. \quad (28)$$

- ② **Essentially cyclic update rule:** There exists a given period $T \geq 1$ during which each index is updated at least once, i.e.

$$\bigcup_{i=1}^T \mathcal{C}^{t+i} = \{0, 1, \dots, K\}, \quad \forall t. \quad (29)$$

We call this update rule a **period- T** essentially cyclic update rule

Extensions to Flexible Update Rules

Algorithm 2. The Flexible ADMM for the Consensus Problem (13)

At each iteration $t + 1$, pick an index set $\mathcal{C}^{t+1} \subseteq \{0, \dots, K\}$.

If $0 \in \mathcal{C}^{t+1}$, compute:

$$x^{t+1} = \arg \min_{x \in X} L(\{x_k^t\}, x; y^t) = \text{prox}_{l(X)+h} \left[\frac{\sum_{k=1}^K \rho_k x_k^t + \sum_{k=1}^K y_k^t}{\sum_{k=1}^K \rho_k} \right].$$

Else $x^{t+1} = x^t$.

If $k \in \mathcal{C}^{t+1}$, node k computes x_k by solving:

$$x_k^{t+1} = \arg \min_{x_k} g_k(x_k) + \langle y_k^t, x_k - x^{t+1} \rangle + \frac{\rho_k}{2} \|x_k - x^{t+1}\|^2.$$

Update the dual variable:

$$y_k^{t+1} = y_k^t + \rho_k (x_k^{t+1} - x^{t+1}).$$

Else $x_k^{t+1} = x_k^t, y_k^{t+1} = y_k^t$.

Extension to Proximal Update Rules

- We can also include proximal update for the x_k subproblems
- **Motivation:** in some applications cheap iterations are preferred

Algorithm 3. A Flexible Proximal ADMM for (13)

At each iteration $t + 1$, compute: $x^{t+1} = \operatorname{argmin}_{x \in X} L(\{x_k^t\}, x; y^t)$.

Pick a set $\mathcal{C}^{t+1} \subseteq \{1, \dots, K\}$.

If $k \in \mathcal{C}^{t+1}$, update x_k by solving:

$$\begin{aligned} x_k^{t+1} = \operatorname{argmin}_{x_k} & \langle \nabla g_k(x^{t+1}), x_k - x^{t+1} \rangle + \langle y_k^t, x_k - x^{t+1} \rangle \\ & + \frac{\rho_k + L_k}{2} \|x_k - x^{t+1}\|^2. \end{aligned}$$

Update the dual variable: $y_k^{t+1} = y_k^t + \rho_k (x_k^{t+1} - x^{t+1})$

Else let $x_k^{t+1} = x_k^t$, $y_k^{t+1} = y_k^t$.

Extension to Proximal Update Rules (cont.)

- **Assumption B.** The stepsize ρ_k is chosen large enough such that:

$$\beta_k := \rho_k - 3L_k - \left(\frac{4L_k}{\rho_k^2} + \frac{1}{\rho_k} \right) 2L_k^2 > 0 \quad (30)$$

$$\alpha_k := \frac{\rho_k}{2} - T^2 \left(\frac{4L_k}{\rho_k^2} + \frac{1}{\rho_k} \right) 8L_k^2 > 0 \quad (31)$$

$$\rho_k \geq 5L_k, \quad k = 1, \dots, K. \quad (32)$$

- **Main Result:** Suppose that Assumptions A1, A3 and B hold. Then the following is true for Algorithm 3.
 - 1 We have $\lim_{t \rightarrow \infty} \|x^{t+1} - x_k^{t+1}\| = 0, k = 1, \dots, K$.
 - 2 Any limit point generated by Algorithm 3 is a stationary solution of problem (13).
 - 3 If X is a compact set, then the sequence converges to the set of stationary solutions

Extension to the Sharing Problem

- Consider the following well-known sharing problem

$$\begin{aligned} \min \quad & f(x_1, \dots, x_K) := \sum_{k=1}^K g_k(x_k) + \ell \left(\sum_{k=1}^K A_k x_k \right) \\ \text{s.t.} \quad & x_k \in X_k, \quad k = 1, \dots, K, \end{aligned} \quad (33)$$

- $x_k \in \mathbb{R}^{N_k}$ is the variable associated with a given agent k
- The variables are coupled through the function $\ell(\cdot)$
- Consider the following reformulation

$$\begin{aligned} \min \quad & \sum_{k=1}^K g_k(x_k) + \ell(\mathbf{x}) \\ \text{s.t.} \quad & \sum_{k=1}^K A_k x_k = \mathbf{x}, \quad x_k \in X_k, \quad k = 1, \dots, K. \end{aligned} \quad (34)$$

- multi-block reformulation**, classical ADMM convergent for convex cases?

Extension to the Sharing Problem (cont.)

Assumption C.

- C1. $\ell(\cdot)$ **Lipschitz continuous** with constant L . Moreover, X_k 's are closed convex sets; each A_k is full column rank, with $\rho_{\min}(A_k^T A_k) > 0$.
 - C2. The stepsize ρ is chosen large enough such that:
 - (1) Each x_k and x subproblem is strongly convex, with modulus $\{\gamma_k(\rho)\}_{k=1}^K$ and $\gamma(\rho)$, respectively.
 - (2) $\rho\gamma(\rho) > 2L^2$, and that $\rho \geq L$.
 - C3. $f(x_1, \dots, x_K)$ is lower bounded over $\prod_{k=1}^K X_k$.
 - C4. g_k is either **smooth nonconvex** or **convex (possibly nonsmooth)**. For the former case, g_k is **Lipschitz continuous** over X_k with constant L_k .
- **Convergence results similar as before.**
 - If the objective is convex, then the stepsize becomes $\rho \geq \sqrt{2}L$

Numerical Results

- We consider a special case of the consensus problem (12)

$$\begin{aligned} \min \quad & \sum_{k=1}^K x^T B_k x + \lambda \|x\|_1 \\ \text{s.t.} \quad & \|x\|_2^2 \leq 1, \end{aligned} \tag{35}$$

- Each step closed form
- $N = 1000$, $K = 10$, $\lambda = 100$. Each $B_k = -\zeta \zeta^H$, where $\zeta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- Each stepsize ρ_k is chosen according to the rule specified in Assumption A2
- We run both the classical and the randomized versions of ADMM, and for the latter case we choose $p_k^t = 0.9$ for all k, t
- We also run the classical ADMM with *small* stepsizes $\hat{\rho}_k = \rho_k / 1000$, $\forall k$

Numerical Results

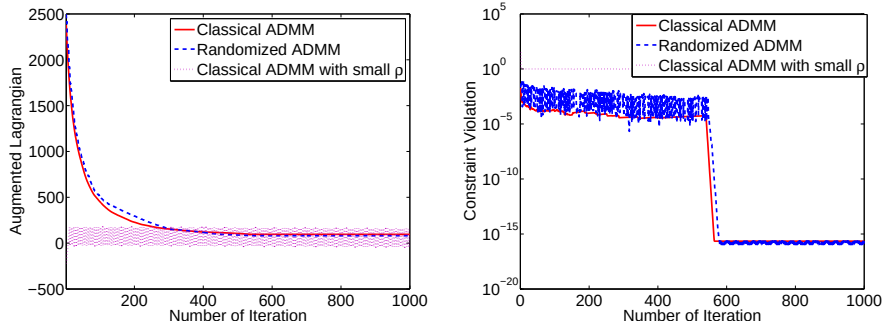


Figure: Top: The value of $L(x^t; y^t)$ for different algorithms. Bottom: The value of $\max_k \{\|x_k^t - x^t\|\}$ for different algorithms.

- Review a few recent applications of ADMM on nonconvex problems
- A new analysis for nonconvex ADMM
 - ① Basic proof steps
 - ② Extensions
- Road map ahead

Weakening the Assumptions

- **Key Questions to Address:** How to relax the three main assumptions
 - ① The Lipschitz continuity assumption on the gradient of the objective function for the **last block** to be updated
- **Example** $\min \|Z - X^T Y\|^2$, s.t. $X = Y$
- Without additional assumptions, the objective of the last subproblem is NOT globally Lipschitz continuous
- **Example** $\min f(x) + \|y\|_1$, s.t. $x = y$, f is nonsmooth as well (e.g., has constraints)

Weakening the Assumptions

- **Key Questions to Address:** How to relax the three main assumptions
 - ① The Lipschitz continuity assumption on the gradient of the objective function for the **last block** to be updated
 - ② Taking into consideration nonconvex constraints
- **Example** Consider $\min \|Y - X_1 X_2\|^2, X_1 \geq 0, X_2 \geq 0$
- Equivalent to

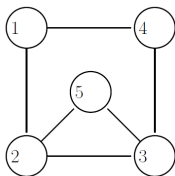
$$\min \|Y - Z\|^2, Z = X_1 X_2, X_1 = \hat{X}_1, X_2 = \hat{X}_2, \hat{X}_1 \geq 0, \hat{X}_2 \geq 0$$

Weakening the Assumptions

- **Key Questions to Address:** How to relax the three main assumptions
 - 1 The Lipschitz continuity assumption on the gradient of the objective function for the **last block** to be updated
 - 2 Taking into consideration nonconvex constraints
 - 3 **General inequality constraints**
- **Example** The consensus problem on a graph (no central controller)

$$\min_k \sum_{k=1}^K g_k(x_k) \text{ s.t. } Ax = 0 \quad (36)$$

where the matrix A is the edge-node incidence matrix



$$A = \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix}.$$

Thank You!